

Project Design Phase-II

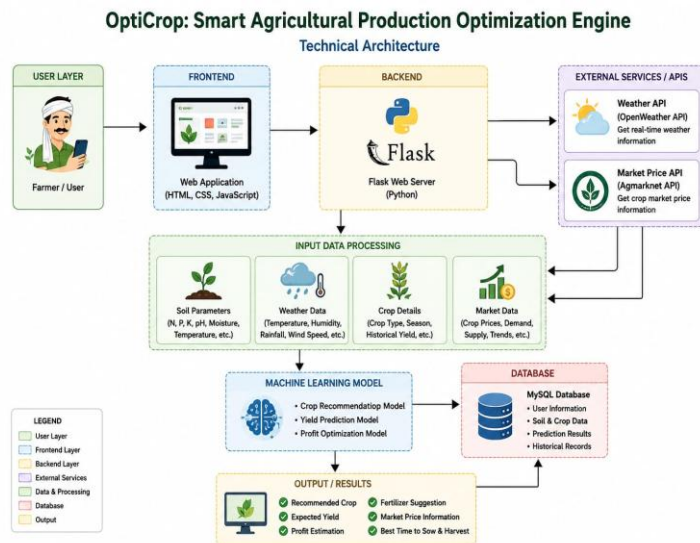
Technology Stack (Architecture & Stack)

Date	July 9 2026
Team ID	SWTID-2026-2791
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	4 Marks

Technical Architecture:

The OptiCrop system helps farmers make informed crop selection decisions by analyzing soil parameters, weather conditions, and historical agricultural data. The system uses a Machine Learning model to recommend the most suitable crop, estimate yield, and improve agricultural productivity. It integrates a web-based user interface, backend processing, database storage, external weather services, and prediction models to deliver accurate recommendations.

References: Flask Documentation <https://flask.palletsprojects.com/>



Guidelines:

- Include all application processes and technology blocks.
- Clearly distinguish between local system and cloud services.
- Show external APIs such as the Weather API.
- Indicate the database and file storage components.
- Include the Machine Learning model and its interaction with the application.
- Display the complete flow from user input to crop recommendation and prediction output

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Farmer enters soil and weather details	HTML, CSS, JavaScript
2.	Application Logic-1	Input Validation and preprocessing	Python ,Flask
3.	Application Logic-2	Crop recommendation and optimization	Scikit-learn,pandas
4.	Application Logic-3	Result generation	Flask
5.	Database	Stores farmer,crop and soil data	MySQL
6.	Cloud Database	Optional cloud storage	Firebase/MongoDB Atlas
7.	File Storage	Dataset and trained model	Local Storage
8.	External API-1	Weather information	OpenWeather API

9.	External API-2	Market price information	Agmarknet API(or simulated API)
10.	Machine Learning Model	Crop recommendation and yield Prediction	Random Forest/XGBoost
11.	Infrastructure (Server / Cloud)	Application Deployment on Local System / Cloud Local Server Configuration: Cloud Server Configuration :	Local PC/Render/AWS

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Interactive web application	HTML, CSS, JavaScript
2.	Security Implementations	User authentication and secure access	Flask Login,HTTPS
3.	Scalable Architecture	Supports multiple farmers	Flask
4.	Availability	Accessible anytime through web	Cloud Deployment
5.	Performance	Fast crop prediction	Scikit-learn

References:

Flask Documentation

<https://flask.palletsprojects.com/>

Scikit-learn Documentation

<https://scikit-learn.org/stable/>

MySQL Documentation

<https://dev.mysql.com/doc/>

Pandas Documentation

<https://pandas.pydata.org/docs/>